

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**DECLARATION OF DR. DAVID K. CHOI
REGARDING CLAIM CONSTRUCTION**

I, David K. Choi, Ph.D., declare as follows:

I. INTRODUCTION AND QUALIFICATIONS

1. I understand that in the above-captioned investigation Complainant GlobalFoundries U.S. Inc. is asserting patent infringement by Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., or Newport Fab LLC (collectively, “Respondents”).

2. I have been asked by Respondents to provide my opinions regarding how a person of ordinary skill in the art (“POSITA”) would have understood certain terms in the asserted claims of U.S. Patents No. 8,507,983 (“’983 patent”). I understand that Complainant is asserting that Respondents have infringed claims 1-7, 13-18, and 20 of the ’983 patent (“asserted claims”).

3. I am being compensated for my work on this matter at my standard hourly rate of \$750 per hour. My compensation is not dependent on the outcome of this case or the content of my opinions or analysis. I do not hold any financial interest in Respondents or Complainant.

4. I reserve the right to supplement or amend this declaration in response to any evidence, opinion, statement, or position taken by Complainant or any expert who submits a declaration in support of Complainant's claim construction positions.

5. I have considered and relied upon my education, background, and experience in forming the opinions expressed in this declaration. In addition, I have reviewed and relied upon the following materials:

- the asserted patents and their prosecution histories;
- the claim terms and proposed constructions of Respondents and Complainant;
- the discovery responses and contentions in the above-captioned case; and
- any other evidence discussed or cited in this declaration, including the evidence listed in Section VII, below.

6. I received my B.S. degree in physics from the University of California, Berkeley in 1993, my M.S. degree in Electrical and Computer Engineering from California State University, Fullerton in 1996, and my Ph.D. degree in Electrical and Computer Engineering from the University of California, Santa Barbara in 2001. By February 2009, I had earned a Ph.D. in Electrical and Computer Engineering and had over 14 years of professional and academic experience in research and product development in digital integrated circuit (IC) design, analog circuit design (including power electronics/power management systems), RF/Microwave circuit design, wireless communications, and semiconductor electronics. I have analyzed and designed numerous high power circuits and systems employing MOS (Metal Oxide Semiconductor), DMOS (Double-diffused MOS), and LDMOS (Laterally Diffused MOS) FET (Field Effect Transistor) transistors since at least 1995.

7. From September 1993 to July 1994, I worked as a research assistant in the Department of Physics at the University of California, Irvine. In this role, I was responsible for the analysis and design of a cryogenic camera system for mid-Infrared imaging of deep space phenomena using (at the time) the largest available (1 Megapixel, 1024x1024) Charge Coupled Device (CCD) and custom Fabry-Perot Interferometer.

8. From January 1995 to August 1996, I worked as a Digital Design Engineer at Western Digital Corporation in the ASIC design group (acquired by Adaptec Inc.). In this role, I used a number of industry standard VLSI design tools from Cadence, Synopsis, Meta, etc. to perform static critical delay analysis, characterize and generate libraries for custom digital logic cells, and to analyze and design clock trees and custom ASIC pad driver cells. I received an award for outstanding achievement for developing a “no re-spin” solution to correct a design error preventing a new ASIC from being powered on and was recognized for my analysis of circuit and system level meta-stable logic.

9. I worked at the Nokia Research Center as a Senior Research Engineer from April 2001 to March 2006. In that role, I designed, built, and tested RF circuits, RF systems, and power management systems. My work included designing a novel LDMOS-based Chireix Outphasing (base station) transmitter for UMTS systems, development of an OFDM RF system simulator (with physical impairments and forward error correction (FEC)), development of baseband PAPR reduction techniques for OFDM radio systems, development of a high power, high precision, 16-bit digitally-controlled (Half-bridge) switched-mode power supply, and development of compact planar antennas on printed circuit substrates for use in mobile phones.

10. From July 2006 to February 2007, I worked as a group leader and staff engineer in the Infrastructure Products Group at RF Micro Devices. In that role, I worked on advanced product

development, including the development of a GaN (Gallium Nitride) HEMT (High Electron Mobility Transistor) dual carrier UMTS Doherty base station transmitter and a scalable GaN HEMT non-linear simulator model.

11. From March 2007 to July 2008, I worked as a Section Manager of the Technology Strategy Team in Technology Intelligence & Collaboration at Samsung Electronics. In that role, I worked on technical feasibility and business opportunity analysis of nanotechnology, and market and technical analysis of biomedical electronics and THz imaging.

12. From July 2008 to February 2011, I worked as a Technical Manager and Principal Engineer in the Advanced Development Group at Aethercomm, Inc. In that role, I worked on product development of pulsed/CW wideband, high power RF transmitters implemented in silicon LDMOS, GaAs (Gallium Arsenide) FET and HBT (Heterojunction Bipolar Transistor), and GaN HEMT device technologies. I also worked on the design of other RF/microwave components such as transformers, baluns, power detectors, switches, filters, magnetics (including inductors with and without ferrites), phase shifters, frequency control, signal processing elements and rectangular waveguides, microwave cavities, as well as, DC power management systems (including full and half-bridge converters), control logic, temperature compensation, fault protection, etc.

13. From May 2011 to December 2012, I worked as a Senior R&D RF Hardware Engineer in R&D Hardware Engineering at LGS Bell Labs Innovations. In that role, I developed high power, highly-linear, and efficient RF transmitter solutions for various wireless standards (including GSM/EDGE, CDMA, WCDMA, and LTE) using peak to average power reduction (PAPR) techniques, analog and digital (pre-distortion) linearization, and various high-Q (SAW, BAW, FBAR, ceramic, and cavity) filters.

14. From August 2013 to March 2014, I worked as a Technical Director in the RF/microwave Power Amplifier Business Unit at Mercury Systems, Inc. In that role, I directed and led all aspects of Mercury Systems' RF/microwave Power Amplifier Business Unit.

15. Since February 2012, I have been the Principal and Owner of RF Design Concepts, Ltd. In this role, I provide consulting services, including product design and development related to RF/microwave circuits, power amplifiers, power management systems (including wireless charging applications), and semiconductor fabrication, advanced IC packaging, as well as for nascent scientific research/technologies.

16. In addition to my work experience in industry, I have also taught electrical engineering (Microelectronics i.e., semiconductor circuits, diodes, bipolar junction and MOS transistors, and transistor circuits) at the university level. From March 2013 to March 2014 and from August 2015 to August 2016, I was an Adjunct Professor in the Department of Electrical, Computer, and Energy Engineering at the University of Colorado, Boulder where I gave graduate level electrical engineering seminars and supported graduate research in wireless communications (including transmitter, combiner, and advanced architectures), wireless charging (including antennas and antenna arrays), and other related topics. Also, from August 2004 to August 2008, I was a Visiting Scholar in the Department of Electrical & Computer Engineering at the University of California, San Diego. I have several published articles in the field of electrical engineering and power amplifier design, including adaptive impedance matching. A list of these publications can be found in my curriculum vitae, which is attached as Exhibit F.

17. My education, employment history, publications, awards, and other relevant information are further detailed in my curriculum vitae ("CV"), a copy of which is provided as Exhibit F to this declaration and which I hereby incorporate by reference.

II. APPLICABLE LEGAL PRINCIPLES

18. I am not a lawyer and am not offering legal opinions. I was asked by counsel for Respondents to provide opinions in the context of certain legal standards summarized below. I rely upon these legal standards as background and to assist in forming my opinions. Counsel has informed me that claim construction is a question of law for the Court to decide in this investigation, potentially with underlying issues of fact. I understand from counsel that the purpose of claim construction is to determine the meaning and scope of the asserted claims from the standpoint of a POSITA.

19. I have been informed by counsel that a disputed claim term should be read not only in the context of the claims, but also in the context of the entire patent-in-suit, including the specification and prosecution history. I understand from counsel that the specification is considered to be the single best guide to the meaning of a disputed term. I understand from counsel that the specification typically supplies the best context for understanding claim meaning. I have been informed by counsel that related patents can provide context for claim construction as well.

20. I understand from counsel that the prosecution histories of patents can additionally inform the meaning of claim language, including by demonstrating how the inventors understood the alleged inventions, whether the inventors limited the alleged inventions during the course of prosecution, and how the examiner understood the claims. I also understand from counsel that the inventor may explicitly or implicitly define a term in the specification of a patent and that this is known as “lexicography.” I have been informed by counsel that, when the meaning of a claim term is not clear from the intrinsic record, the court may also consult extrinsic evidence, such as expert testimony, in construing the term.

21. I understand from counsel that claim construction must be understood from the standpoint of a POSITA at the time of alleged invention. I also understand from counsel that in determining who would have been a POSITA, it is proper to consider the types of problems encountered in the art, prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the education level of active workers in the field. I have provided my opinion below as to who in my opinion would have been a POSITA in the field of the asserted patents.

22. Finally, I understand from counsel that a patent claim is invalid as indefinite if the claim, as read in light of the specification delineating the patent, and the prosecution history, fails to inform, with reasonable certainty, those skilled in the art about the scope of the invention. I have been informed by counsel that a claim term does not become reasonably certain simply because a skilled artisan, when pressed, might manage to articulate a definition for it.

III. PERSON OF ORDINARY SKILL IN THE ART

23. As noted above, I understand from counsel that the asserted claims must be construed from the point of view of a POSITA as of the earliest claimed priority date. For the '983 patent, I have assumed the priority date to be February 23, 2009, which is the filing date of the earliest application to which the '983 patent claims priority. I have been informed by counsel that Complainant does not allege that the '983 patent is entitled to a priority date earlier than February 23, 2009.

24. The relevant technical field for the asserted patents is the design and fabrication of semiconductor devices, including the formation of laterally diffused metal oxide semiconductor (LDMOS) devices. In my opinion, based on the legal principles I discussed above, my experience in and knowledge of the field and those who work and study in it, and the materials I have

considered in preparing this declaration, a POSITA in this technical field on February 23, 2009, would have had at least a Bachelor of Science in electrical engineering, materials science, materials engineering, materials science and engineering, chemical engineering, or similar majors with 2-3 years of experience related to the design and fabrication of semiconductor devices. More education could substitute for experience or vice versa.

25. My opinions in this declaration would not change if the POSITA were determined to have somewhat more or less or slightly different education and/or experience than what I have summarized above. My knowledge and skill exceed that of a POSITA. My opinions herein reflect and are made from the perspective of a POSITA as of February 23, 2009.

IV. THE '983 PATENT

26. As noted above, I have been informed by counsel that Complainant is asserting claims 1-7, 13-18, and 20 of the '983 patent. I list the asserted claims below.

A. Asserted Claims of the '983 Patent

27. Asserted claims 1-7, 13-18, and 20 of the '983 patent recite as follows:

1[p] A device comprising:

1[a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

1[b] a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and

1[c] a doped drift well adjacent to a second edge of the gate.

2. The device of claim 1 further comprising spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

3[p] The device of claim 1 wherein the substrate comprises:

3[a] a drift isolation region in the doped drift well; and

3[b] a doped deep well in the substrate of the active device region encompassing the doped drift well.

4. The device of claim 3 wherein the drift isolation region is disposed under the gate.

5[p] The device of claim 4 wherein the first and a second heavily doped regions are provided adjacent to the gate, wherein:

5[a] the first heavily doped region is disposed in the doped channel well on a source side of the device; and

5[b] the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

6. The device of claim 1 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

7. The device of claim 1 wherein the channel having a channel length that is controlled by ion implantation.

13[p] A device comprising:

13[a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

13[b] a doped channel well disposed in the substrate adjacent to a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate;

13[c] a doped drift well adjacent to a second edge of the gate; and

13[d] a drift isolation region in the doped drift well.

14. The device of claim 13 further comprising a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

15. The device of claim 13 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

16. The device of claim 13 wherein the drift isolation region is disposed under the gate.

17[p] The device of claim 13 wherein the first and a second heavily doped regions are provided adjacent to the gate, wherein:

17[a] the first heavily doped region is disposed in the doped channel well on a source side of the device; and

17[b] the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

18. The device of claim 13 wherein the channel having a channel length that is controlled by ion implantation.

20[p] A semiconductor device comprising:

20[a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

20[b] first and second heavily doped regions adjacent to first and second edges of the gate;

20[c] a doped channel well disposed in the substrate adjacent to the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from the first heavily doped region adjacent to the first edge of the gate; and

20[d] a doped drift well adjacent to a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well.

B. The '983 Patent

28. The '983 patent is titled "High Voltage Device." It lists two named inventors: Guowei Zhang and Purakh Raj Verma.

29. The '983 patent discloses that "Lateral Double-Diffused (LD) transistors have been widely employed in high voltage applications." ('983 patent, 1:16-17.) "For high performance

LD transistors, low drain-to-source on-resistance ($R_{ds_{on}}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability.” (*Id.*, 1:17-21.) The ’983 patent discloses that, “[t]o achieve low $R_{ds_{on}}$, the channel of the high LD transistor should be as short as possible.” (*Id.*, 1:21-22.)

30. The ’983 patent also discloses that, “[a]s technology evolves into era of deep sub-micron (e.g., beyond 0.35 μm) Very-Large-Scale Integration (VLSI), there is a desire for both high voltage (HV) LD transistors and low voltage (LV) transistors to be fabricated on the same substrate.” (*Id.*, 1:23-27.) The ’983 patent discloses that complementary metal oxide semiconductor (CMOS) processes are “[g]enerally...used to form” LV and HV devices. (*Id.*, 1:27-29.) The ’983 patent discloses that “conventional CMOS processes are incompatible for forming HV transistors.” (*Id.*, 1:29-31.) The ’983 patent discloses:

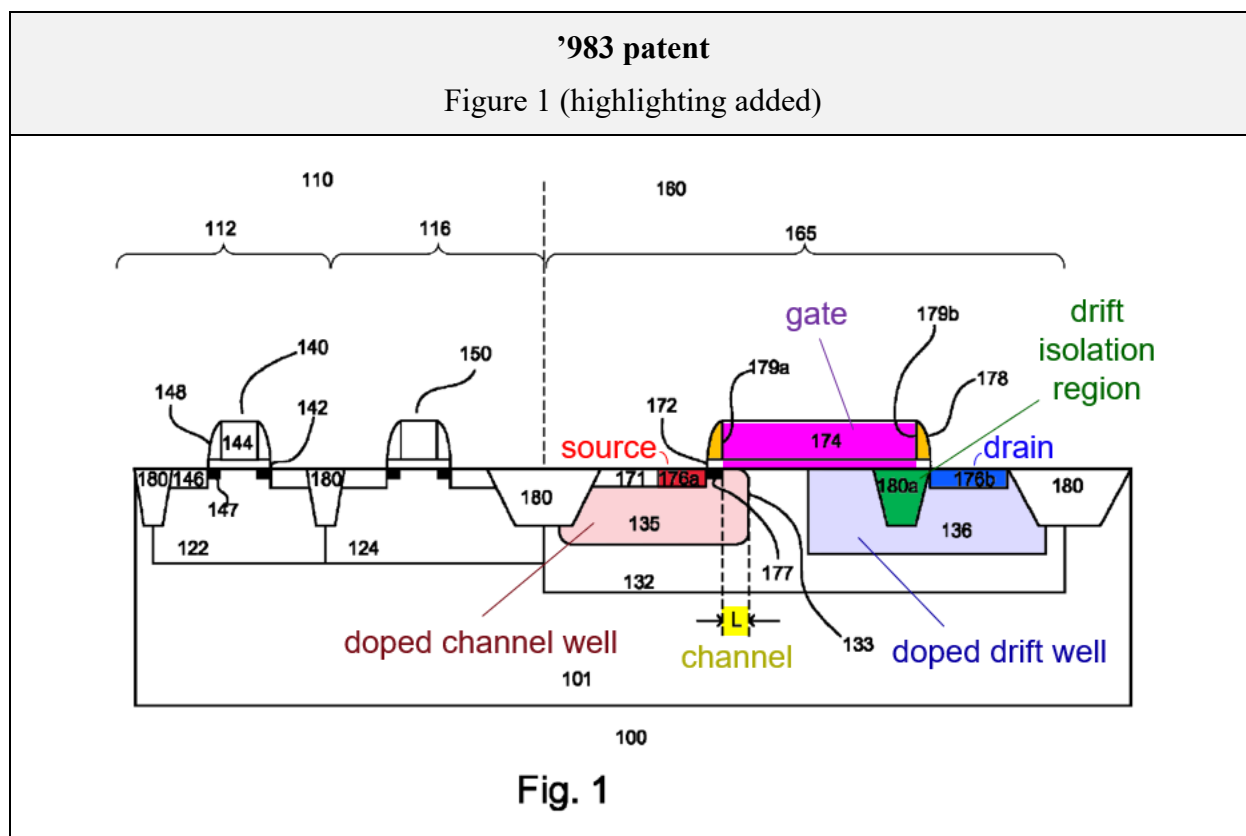
For example, process overlay issues make it difficult for alignment of the HV channel well, which defines the channel length. Process variations in process overlay requires large process windows. This makes forming short channel lengths to achieve low $R_{ds_{on}}$ difficult if not impossible. Furthermore, the thin gate electrodes used prevent them serving as a hard mask, further exacerbating process control in forming the channel well.

(*Id.*, 1:31-38.) The ’983 patent discloses that “it is desirable to provide reliable HV devices formed with short channel lengths to have low $R_{ds_{on}}$.” (*Id.*, 1:39-41.)

31. The ’983 patent discloses embodiments that “generally relate to semiconductor devices or integrated circuits.” (’983 patent, 2:43-44.) “More particularly, some embodiments relate to high power devices,” including “lateral double-diffused (LD) transistors, such as metal oxide transistors (MOS).” (*Id.*, 2:44-47.)

32. Figure 1 of the ’983 patent shows a substrate **101** with two regions (**110** and **160**), where, “[i]n one embodiment, the first region comprises a low voltage (LV) device region while

the second region comprises a high voltage (HV) device region.” (’983 patent, 2:61-64.) Figure 1 is reproduced below with highlighting that I explain further below:



33. The HV region **160** comprises a **gate**, which “includes a HV gate electrode **174** over a HV gate dielectric **172**.” (’983 patent, 4:49-50.) The gate includes “first and second edges **179 a-b** or sidewalls.” (*Id.*, 4:46-48.) “Sidewalls of the gate can be provided with **sidewall spacers 178**.” (*Id.*, 5:3-4.) Further, the HV device includes a “**source extension region 177**,” which “extends under the spacer on the channel side of the HV gate.” (*Id.*, 5:30-32.)

34. The HV device in Figure 1 further includes a **source diffusion region 176a** and a **drain diffusion region 176b**. (’983 patent, 5:25-30.) The source diffusion region **176a** is “provided in the **channel well 135**,” which the specification refers to as a “channel (or body) well” or “implant.” (*Id.*, 5:28-30, 4:17-19, 5:62-65.) “An edge **133** of the channel well **135** extends below

the gate on the source side of the gate.” (*Id.*, 5:51-52.) “The channel edge of the channel well **135** and the first edge **179a** of the gate on the source side define a **channel** of the HV transistor having an effective channel length L.” (*Id.*, 5:53-56.)

35. The HV device in Figure 1 also includes a **drift well 136**. (*Id.*, 4:17-19.) The ’983 patent discloses that “[t]he HV drift well, for example, defines the type of HV device. For example, a p-type HV active region would have a n-type HV channel well and a p-type HV drift well.” (*Id.*, 4:20-22.)

36. The ’983 patent further discloses that a “**drift isolation region 180a** is disposed in the drift well **136** under the gate. Disposing the drift well at other positions, such as partially underlying the gate may also useful.” (*Id.*, 5:21-24.)

37. Other structures in Figure 1 include “[i]solation regions **180**, such as shallow trench isolation (STI) regions are provided to isolate the LV and HV regions as well as active sub-regions within the regions.” (’983 patent, 4:11-13.) Also, the ’983 patent indicates that “[a] deep well **132** may be provided in the HV active region, encompassing the HV channel and drift wells.” (*Id.*, 4:27-28.) “Generally, the deep well comprises dopants of the first type or same as the HV device type.” (*Id.*, 4:28-30.) Further, the ’983 patent discloses a “body contact region **171**,” which provides electrical coupling to the channel well **135**. (*Id.*, 5:49-50.)

V. MATERIALS CONSIDERED

38. In arriving at my opinions in this declaration, I have considered the asserted patents, their file histories, the parties’ proposed claim constructions, and all other information and documents cited or discussed in this declaration, including the exhibits below.

Exhibit No.	Description
N/A	Complaint, <i>Certain Semiconductor Devices, Products Containing Same, and Components Thereof</i> , Investigation No. 337-TA-1500 (“Complaint”), and accompanying exhibits and appendices
N/A	Certified copy of U.S. Patent No. 8,507,983 (Exhibit 2 to Complaint) (“’983 patent”)
N/A	Certified copy of Prosecution History of the U.S. Patent No. 8,507,983 (Appendix B to Complaint)
21	U.S. Patent Application Publication No. 2008/0191291A1 (“Wu ’291”) (TOWER_1500_00034458)
22	U.S. Patent Application Publication No. 2009/0020811A1 (“Voldman ’811”) (TOWER_1500_00034410)
23	This exhibit was intentionally left blank
24	’983 patent prosecution history: July 16, 2012 Application (GF-ITC-00004497 – 4571)
25	U.S. Patent No. 6,979,864 (“Negoro”)
26	U.S. Patent No. 7,935,992 (“Chang”)
27	’983 patent prosecution history: Jan. 25, 2013 Amendment (GF-ITC-00004596 – 4610)
28	’983 patent prosecution history: October 26, 2012 Non-Final Rejection (GF-ITC-00004576 – 4593)
29	U.S. Patent No. 8,053,319 (“Liu”)
30	U.S. Patent No. 8,222,130 (“Zhang”)
31	Excerpt from <i>Micro-Electronic Circuits</i> , Adel S. Sedra et al., CBS College Publishing (1st Ed.; 1982) (TOWER_1500_00040579)
32	Excerpt from Complainant GlobalFoundries U.S. Inc.’s Second Supplemental Objections and Responses to Respondents’ First Set of Interrogatories (Nos. 1-73) (“GF validity contentions”)
33	Excerpt from <i>Micro-Electronic Circuits</i> , Adel S. Sedra et al., Saunders College Publishing (3d Ed.; 1991)

VI. CLAIM TERMS AND PROPOSED CONSTRUCTIONS

39. I understand that the parties have proposed the following claim constructions for the following disputed terms in the ’983 patent:

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“adjacent to” ’983 patent, claims 1, 5, 13, 17, and 20	Plain and ordinary meaning; not indefinite.	Indefinite
“the channel is displaced from [a/the] first heavily	Plain and ordinary meaning; not indefinite.	Indefinite

Disputed Claim Term	Complainant's Proposed Construction	Respondents' Proposed Construction
doped region adjacent to the first edge of the gate" '983 patent, claims 1, 13, and 20		
"the drift isolation region is disposed under the gate" '983 patent, claims 4 and 16	Plain and ordinary meaning	"the drift isolation region is positioned at least in part under the gate"
"the channel having a channel length that is controlled by ion implantation" '983 patent, claims 7 and 18	Plain and ordinary meaning	"the channel having a channel length that is formed by implanting ions into the substrate to form a channel well"
"edge of the gate" '983 patent, claims 1, 2, 6, 13, 14, 15, and 20	Plain and ordinary meaning	"edge of the gate electrode over the gate dielectric"

40. I have been informed by counsel that the parties have not agreed upon a construction for any claim term in the '983 patent. Below, I provide opinions about the parties' proposed constructions for disputed claim terms for the '983 patent. I understand from counsel that Respondents dispute indefiniteness and claim construction for other terms in the '983 patent but that, to narrow disputes, only the above five terms have been identified for claim construction briefing at this time. I have been asked to provide my opinions regarding certain issues related to some of the disputed claim terms for the '983 patent. I reserve the right to address other disputed issues of indefiniteness and claim construction.

VII. CLAIM CONSTRUCTION

A. “adjacent to” / “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“adjacent to” '983 patent, claims 1, 5, 13, 17, and 20	Plain and ordinary meaning; not indefinite.	Indefinite
“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” '983 patent, claims 1, 13, and 20	Plain and ordinary meaning; not indefinite.	Indefinite

41. In my opinion, the asserted claims of the '983 patent are indefinite because the terms “adjacent to” and “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” fail to inform a POSITA about the scope of the alleged inventions with reasonable certainty, when read in light of the patents’ specifications and prosecution histories.

42. The term “adjacent to” appears throughout the asserted claims, as shown below:

1[p] A device comprising:

1[a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

1[b] a doped channel well disposed in the substrate **adjacent to** a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region **adjacent to** the first edge of the gate; and

1[c] a doped drift well **adjacent to** a second edge of the gate.

5[p] The device of claim 4 wherein the first and a second heavily doped regions are provided **adjacent to** the gate, wherein:

5[a] the first heavily doped region is disposed in the doped channel well on a source side of the device; and

5[b] the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

13[p] A device comprising:

13[a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

13[b] a doped channel well disposed in the substrate **adjacent to** a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region **adjacent to** the first edge of the gate;

13[c] a doped drift well **adjacent to** a second edge of the gate; and

13[d] a drift isolation region in the doped drift well.

17[p] The device of claim 13 wherein the first and a second heavily doped regions are provided **adjacent to** the gate, wherein:

17[a] the first heavily doped region is disposed in the doped channel well on a source side of the device; and

17[b] the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

20[p] A semiconductor device comprising:

20[a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

20[b] first and second heavily doped regions **adjacent to** first and second edges of the gate;

20[c] a doped channel well disposed in the substrate **adjacent to** the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from the first heavily doped region **adjacent to** the first edge of the gate; and

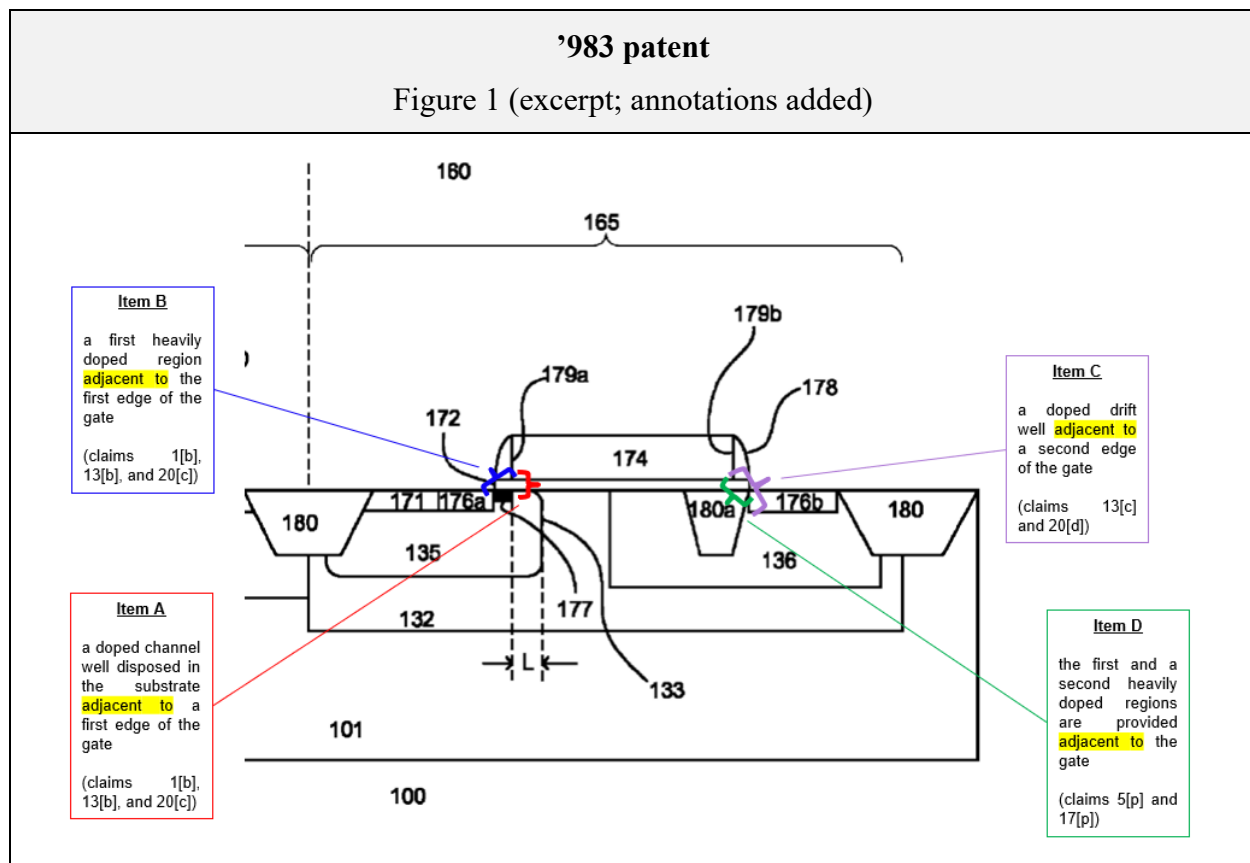
20[d] a doped drift well **adjacent to** a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well.

43. Below is a list of the elements that the claims require to be “adjacent to” each other.
- a. **Item A:** the “doped channel well” must be “adjacent to” the “first edge of the gate” (claims 1[b], 13[b], and 20[c]);
 - b. **Item B:** the “first heavily doped region” must be “adjacent to” the “first edge of the gate” (claims 1[b], 13[b], and 20[c]);
 - c. **Item C:** the “doped drift well” must be “adjacent to” the “second edge of the gate” (claims 13[c], 20[d]); and
 - d. **Item D:** the “first heavily doped region” and “second heavily doped region” must be “adjacent to” the “gate” (claims 5[p] and 17[p]).

44. It is my opinion that a POSITA would not have had reasonable certainty as to what distance or proximity is required to satisfy the claimed “adjacent to” relationships in the claims. It is my opinion that it would have been especially unclear to a POSITA how each of the claimed “adjacent” elements could be understood to be “adjacent to” each other despite wide variances in the ways they are required to be configured with respect to one another in Items A-D, above. Each of the claimed elements describe different spatial relationships yet use the same “adjacent to” claim

term. For instance, in the list above, Item A requires the “first edge of the gate” to be “adjacent to” the “doped channel well” while Item B requires the “first edge of the gate” to be “adjacent to” “the first heavily doped region,” even though the “doped channel well” and “the first heavily doped region” have different spatial and proximal relationships to the “first edge of the gate.”

45. The following demonstrative diagram shows the elements in Figure 1 of the '983 patent that need to be “adjacent to” each other according to example '983 patent claims:



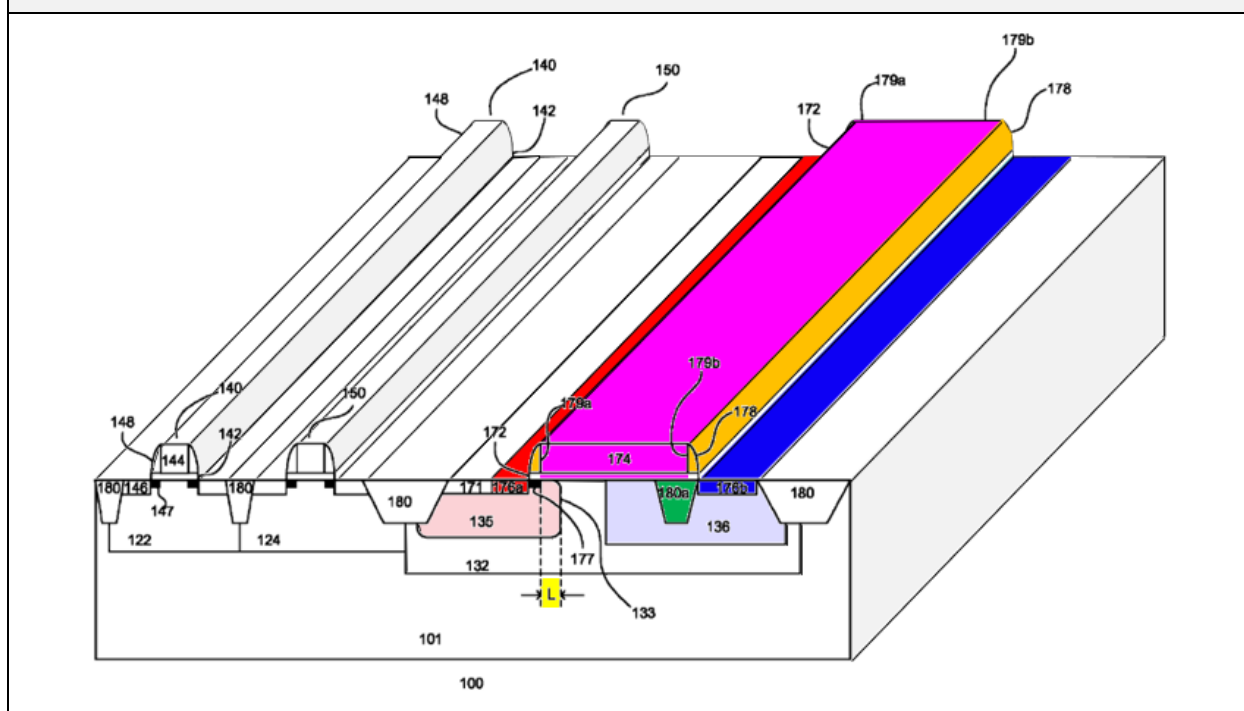
The above diagram shows that the same “adjacent to” claim language is used inconsistently, claiming a certain spatial relationship between elements in one instance and a different spatial relationship in another. Moreover, the same “adjacent to” claim language refers to distances of different magnitudes and distances that extend through various interceding structural elements. In

my opinion, a POSITA would not have had reasonable certainty as to the meaning or scope of the “adjacent to” claim language, particularly in light of the diverse spatial relationships that the claim language is required to cover.

46. Claims 1, 13, and 20 of the '983 patent require a “channel [] displaced from a first heavily doped region adjacent to the first edge of the gate.” A POSITA would not have had reasonable certainty in terms of the scope of spatial relationships encompassed by that claim language. For example, a POSITA would not have had any basis for understanding what is the limit outside of which the first heavily doped region is no longer “adjacent to” the first edge of the gate. Neither “the first edge of the gate” nor “a first heavily doped region” refer to the necessary referent points in space a POSITA would need to determine whether or not these structures are “adjacent to” one another. Quite the contrary. The “first edge of the gate,” for example, in Figure 1, is depicted as a line (which is also not a point in space). A POSITA would have understood that, that line depicting “the first edge of the gate,” would actually extend out of and into the page because a POSITA would have understood that Figure 1 depicts a cross-sectional view of a three-dimensional device,¹ as shown below:

¹ See, e.g., Ex. 31 at 300 (Figure 5.1 showing three-dimensional MOSFET device).

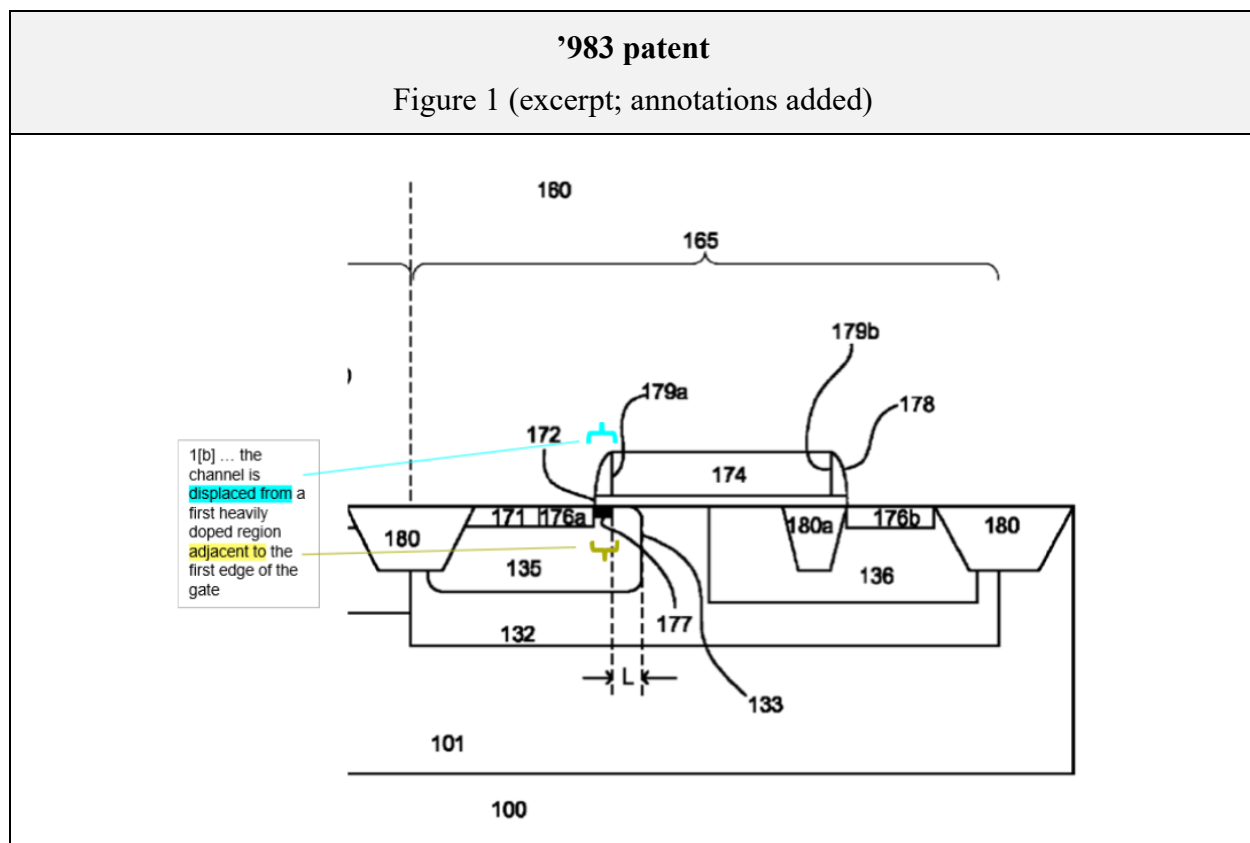
'983 patent
Figure 1 (modified)



In the three-dimensional world in which these devices actually exist, a POSITA would not have had any reasonable certainty in terms of what point to measure the claimed “adjacen[cy]” relationship when looking at the respective three-dimensional structures. Given this reality, a POSITA would not have understood whether the term “adjacent to” would be satisfied by a picometer, a nanometer, a micrometer (microns), 10’s of microns, 100’s microns, etc.

47. A POSITA would have particularly lacked any objective guidance in the context of LDMOS devices, since such devices are laterally diffused and therefore can have significant variation in terms of the distances between laterally distributed structures in the device. A POSITA would also not have understood how many interceding structural elements could be situated between two “adjacent” elements—*e.g.*, one, two, three, etc. As a result, a POSITA would not have had any reasonable certainty for judging whether a particular device has, for example, a “first

heavily doped region” and a “first edge of the gate” located such that those two structures would be considered “adjacent to” each other within the scope of the ’983 patent claims.

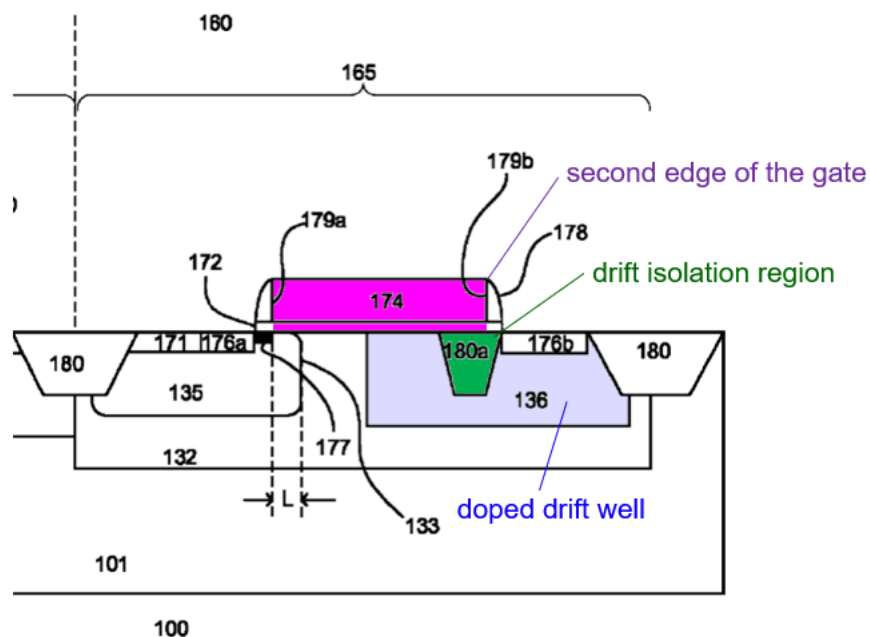


49. A POSITA would not have understood with reasonable certainty how this claim element, which jointly requires that the “first heavily doped region” be both “adjacent to” *and* “displaced from” the “first edge of the gate,” would be satisfied when the “adjacent to” and “displaced from” are ostensibly conflicting requirements. Faced with this language, a POSITA would have had no reasonable certainty in understanding which devices containing these structures fall within, and which devices fall outside of, the range of distances in the scope of the contradictory “adjacent to” and “displaced from” claim language.

50. As another example, claims 1, 13, and 20 require a “doped drift well” that is “adjacent to” the “second edge of the gate.” But in the embodiments, there is a “drift isolation region” in the “doped drift well.” For example, Figure 1 of the ’983 patent shows that this “drift isolation region” can be in between the “doped drift well” and the “second edge of the gate.” Further, the specification explains that “drift isolation region 180a may be provided within the drift well 136 on the high voltage side of the HV device. The drift isolation region 180a can improve breakdown voltage of the HV device due to high voltage. For example, the drift isolation region is particularly useful to sustain high voltages for applications beyond about 10V. As shown, the drift isolation region 180a is disposed in the drift well 136 under the gate.” (’983 patent, 5:15-22).

'983 patent

Figure 1 (excerpt; annotations added)

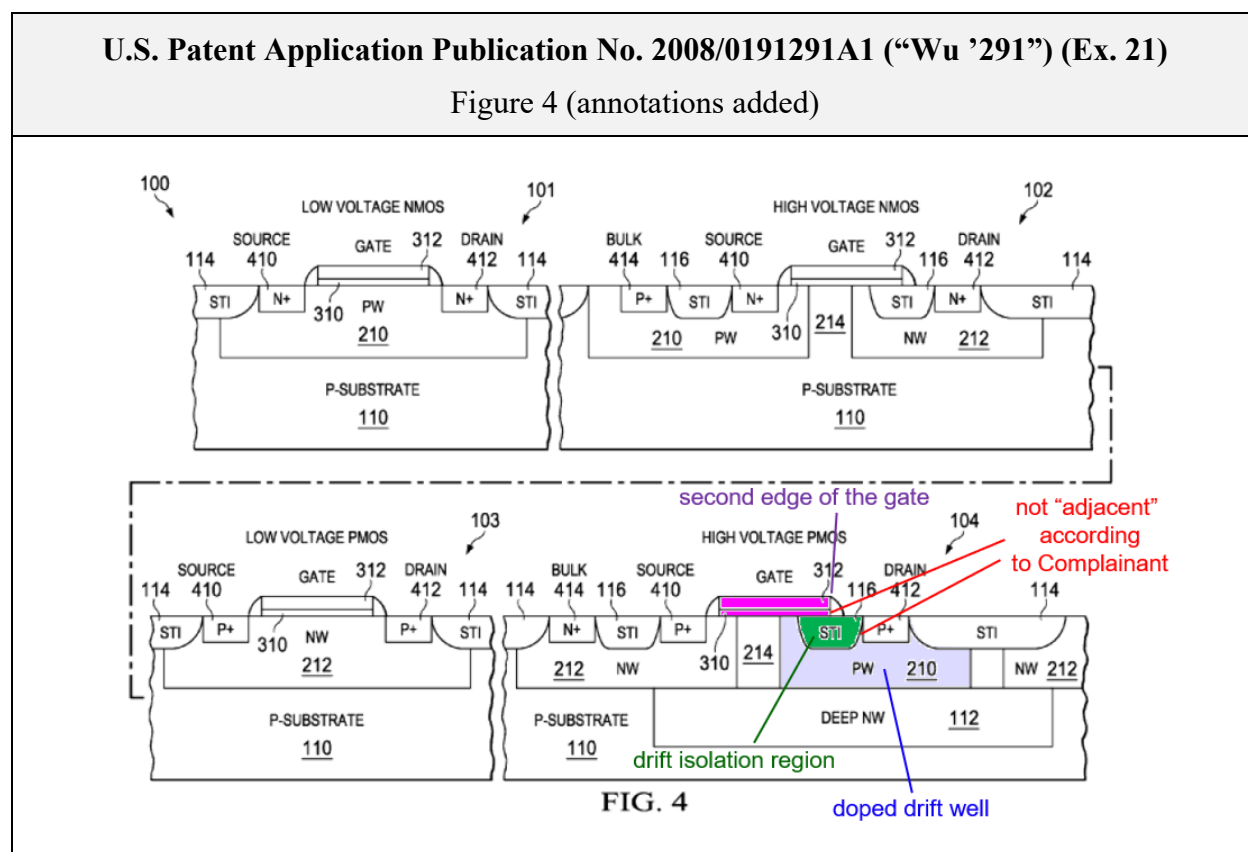


Accordingly, the spatial relationship between the doped drift well and the second edge of the gate may vary based on the spatial dimensions of the drift isolation region. A POSITA would have understood, and the specification of the '983 patent confirms, that such spatial dimensions and relationships are highly dependent on the design of a given device and its required specifications and application. (*E.g.*, '983 patent, 5:15-24 (“Disposing the drift well at other positions, such as partially underlying the gate may also useful.”), 7:8-13.)

51. The lack of reasonable certainty in a POSITA’s understanding of the “adjacent to” term is particularly evident given the fact that Complainant is arbitrarily and without explanation arguing that certain elements in the prior art are not “adjacent to” each other.

52. For example, Complainant raises such an argument for the prior art reference U.S. Patent Application Publication No. 2008/0191291A1 (“Wu ’291”) (Ex. 21). Complainant

argues that “Tower has not established the asserted adjacency relationship between the drift well and the gate’s second edge from Wu ’291’s cited passages.” (Ex. 32 at 66.) The below demonstrative diagram shows the spatial relationship Complainant is arguing does not satisfy the “adjacent to” claim language. A POSITA would have had no reasonable certainty as to why this spatial relationship does not represent two elements (in this case “second edge of the gate” and the “drift well”) that are “adjacent to” each other, and Complainant offers no rationale to support its position. Complainant’s own argument about Wu ’291 again demonstrates the lack of reasonable certainty as to the scope of the “adjacent to” claim language.



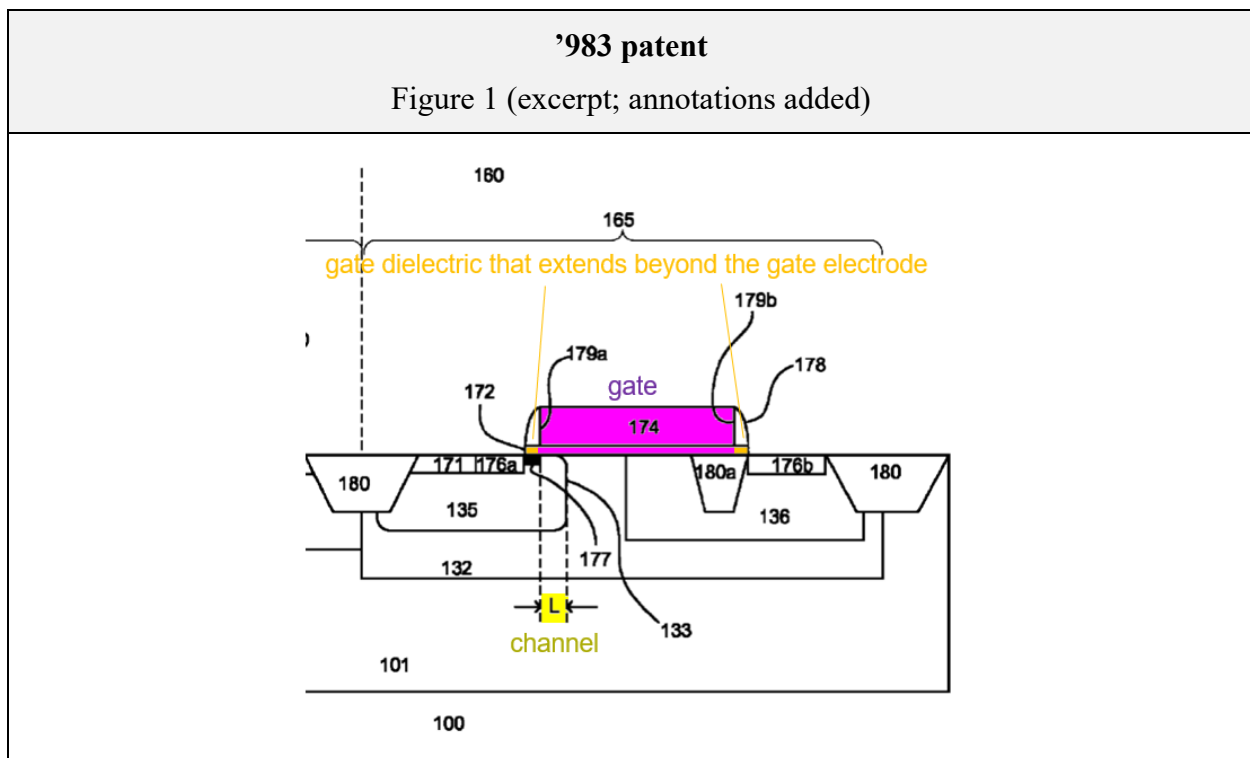
53. As another example, Complainant raises a dispute as to the “adjacent to” claim language for the prior art reference U.S. Patent Application Publication No. 2009/0020811A1

B. “edge of the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“edge of the gate” ’983 patent, claims 1, 2, 6, 13, 14, 15, and 20	Plain and ordinary meaning	“edge of the gate electrode over the gate dielectric”

54. It is my opinion that the term “edge of the gate” should be construed as “edge of the gate electrode over the gate dielectric.”

55. The claims of the ’983 patent indicate that “the first edge of the gate and channel edge beneath the gate define a channel of the device[.]” A POSITA would have understood that this claim language must mean that the “first edge of the gate” is the edge of the gate electrode over the gate dielectric. A POSITA would have understood that any other interpretation would be nonsensical. For example, Figure 1 of the ’983 patent shows a section of dielectric that extends beyond the gate electrode, as shown below:



Looking at annotated Figure 1 of the '983 patent, above, where “L” denotes the channel, and, for instance, the language of claims 1 (“the first edge of the gate and channel edge beneath the gate define a channel of the device”) and 6 (“the first edge of the gate corresponds to a channel edge of the gate of the device”), a POSITA would have understood that the channel does not extend to the end of the gate dielectric beyond the gate electrode—*i.e.* where there is no gate electrode. ('983 patent, 13:23-24, 13:45-46). A POSITA would have understood that a “channel” can only exist where there is a gate electrode over a gate dielectric, since the formation of a channel in a metal oxide semiconductor field effect transistor (MOSFET) device, such as what the '983 patent discloses, depends on the effect of the electric field produced by the voltage on the gate electrode, which is electrically isolated from the underlying semiconductor by the gate dielectric. Accordingly, a POSITA would have understood that a channel is formed where a gate electrode is over a gate dielectric. A POSITA would have understood that a channel does not exist where there is only, for instance, gate dielectric but no gate electrode above it.

56. Further, it is my view that, in the context of a MOSFET device, a POSITA would have understood a “gate” to be a structure that exists only where there is an electrode over a dielectric. A gate is a well-established concept in MOSFETs, where a “thin oxide layer is grown on the surface of the channel, and metal (aluminum) is deposited on it to form the gate (G). It should be clear that the name MOS is derived from the structure of the device.” (Ex. 31 at 303 (TOWER_1500_00040897)). The figure below shows the “Gate (G),” gate “Oxide,” and the gate electrode (depicted as a dark line) overlying the gate oxide.

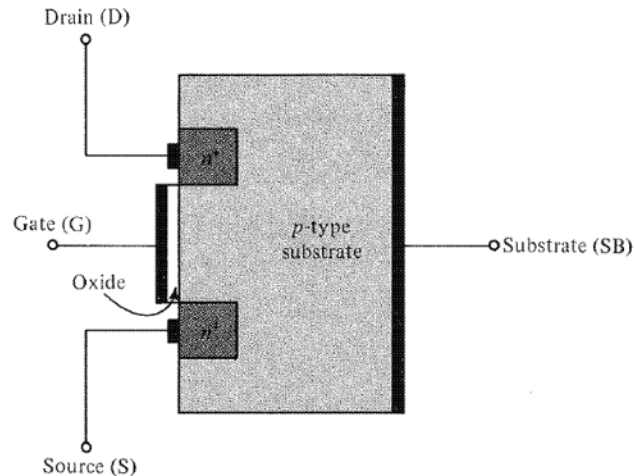


Fig. 8.8 Physical structure of n-channel enhancement-type MOSFET.

In a MOSFET, the gate refers to an electrode that can control the electrical conduction of the underlying semiconductor substrate depending on the voltage applied between the gate and the source, so as to turn the MOSFET ON (in a conducting state) or OFF (in a non-conducting state). The gate electrode is electrically isolated (and physically separated) from the substrate by an insulating material, known as a dielectric, so the voltage applied to the gate electrode produces an electric field that attracts or repels charge carriers in the underlying semiconductor substrate (hence the name, “Field Effect Transistor,” in “MOSFET”). The applied gate voltage “enhances” a conductive channel or causes a “depleted” non-conductive channel. A POSITA would have understood that both the electrode and the underlying dielectric are necessary for the operation of a MOSFET and that the current flowing through a MOSFET depends directly on the geometry of

the gate electrode overlying the gate oxide and the gate oxide's electrical properties, as confirmed by the following diagram from a well-established textbook.²

Ex. 33, Excerpt from *Micro-Electronic Circuits*, Adel S. Sedra et al., Saunders College Publishing (3d Ed.; 1991)

p. 300

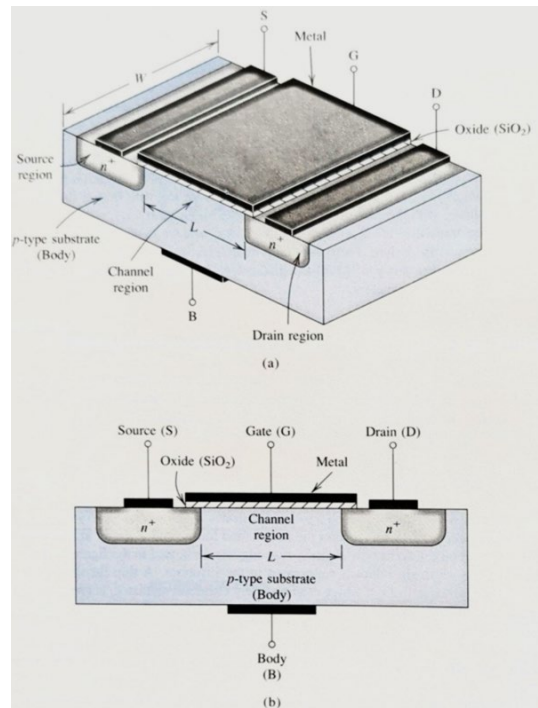


Fig. 5.1 Physical structure of enhancement-type NMOS transistor: (a) perspective view; (b) cross section. Typically $L = 1$ to $10\ \mu\text{m}$, $W = 2$ to $500\ \mu\text{m}$, and the thickness of the oxide layer is of the order of $0.1\ \mu\text{m}$.

² Equation 8.3 of the Sedra and Smith textbook (1982; Ex. 31) gives the generalized MOSFET current equation as, $i_D = \beta[(v_{GS} - V_T)v_{DS} - \frac{1}{2}v_{DS}^2]$, where “ β depends, among other things, on the geometry of the device.” (Ex. 31 at 309 (TOWER_1500_00040896). A later edition of the same textbook (Ex. 33) expresses the generalized MOSFET current equation more explicitly as, $i_D = K[2(v_{GS} - V_T)v_{DS} - v_{DS}^2]$, in which $K = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)$, “ μ_n is a physical constant known as the electron mobility,” “ C_{ox} (called the oxide capacitance) is the capacitance per unit area of the gate-to-body capacitor for which the oxide layer serves as dielectric, L is the length of the channel, and W is its width (refer to Fig. 5.1).” (Ex. 33 at 309-310).

This understanding would have informed a POSITA's understanding that the "edge of the gate" in the asserted claims of the '983 patent is the "edge of the gate electrode over the gate dielectric."

I declare under penalty of perjury that the foregoing statements are true and accurate.

Dated: August 25, 2026



David K. Choi